A single colour-class lower bound for the first open case of the Erdős–Gyárfás problem

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Abstract

The first open case of the Erdős–Gyárfás generalized Ramsey problem [1] asks whether K_{26} admits a *balanced* 5-colouring: one in which every 6 vertices see all 5 colours on their 15 pairs. A counting argument reduces a proof of impossibility to the single colour-class bound $m^* \geq 66$, where m^* is the least number of edges of a graph on 26 vertices with independence number at most 5 in which every 6 vertices span at most 11 edges. We prove $m^* \geq 63$, unconditionally, improving the elementary Turán bound $m^* \geq 55$ by eight. The proof is a finite Füredi-stability "rung" argument whose case analysis is discharged by a full-specification constraint (SAT/CP) decision model, independently validated. A closing section localizes the remaining gap to $m^* \geq 66$ and shows that it provably resists the cheap (fractional and clique) methods, so that closing it is a matter of computation rather than of a missing idea.

1 The problem and the reduction

Erdős and Gyárfás [1] ask, for $r \geq 3$: if the edges of K_{r^2+1} are r-coloured, must some $r+1$ vertices induce a K_{r+1} missing a colour? The cases $r = 3, 4$ are theirs; $r \geq 5$ is open. The first open case is $r = 5$, on K_{26}. Call a 5-colouring *balanced* if every 6 vertices see all 5 colours on their 15 pairs; the conjecture is that none exists.

Fix a colour c and let $G = G_c$ be its colour class. Restricted to one class, balancedness says exactly two things:

(A) $\alpha(G) \leq 5$: every 6-set carries a c-edge, so there is no independent 6-set;

(B) every 6-set spans at most 11 edges of colour c, leaving room for the other four colours on its 15 pairs.

Write m^* for the minimum of $\mathrm{e}(G)$ over all graphs G on 26 vertices satisfying (A) and (B). The five colour classes partition the $\binom{26}{2} = 325$ edges of K_{26}, so if

$$m^* \geq 66 \quad \Longrightarrow \quad 5 \cdot 66 = 330 > 325,$$

no balanced colouring can exist and the $r = 5$ case is settled. The naive bound is $m^* \geq 55$ (Turán applied to the complement). Our result is the following eight-edge improvement.

Theorem 1. *In any balanced 5-colouring of K_{26}, every colour class has at least 63 edges; that is, $m^* \geq 63$.*

*`william.blair0708@gmail.com`. Code, certificates, and the finite decision model are public at https://github.com/willblair0708/verified-combinatorics/tree/076cb35b37ff7d91efe8fc7ff1826d5d898d1ea1/erdos-617.

The bound is unconditional, resting only on Füredi's stability theorem [2] and a finite, machine-checked case analysis (Sections 2 and 3). It falls four short of the value $m^* \geq 66$ that would settle the case; Section 5 explains precisely what the remaining gap requires.

2 The complement and the rung method

Pass to the complement $H = \overline{G}$ on the same 26 vertices. Conditions (A) and (B) become:

- H is K_6-free (a 6-clique in H is an independent 6-set in G);
- every 6-set spans *at least* 4 edges in H (since $15 - 11 = 4$).

As $\mathrm{e}(G) + \mathrm{e}(H) = 325$, proving $m^* \geq 63$ is proving $\mathrm{e}(H) \leq 262$ for every such H.

Suppose $\mathrm{e}(H) = 270 - q$. By Füredi's theorem ([2], Theorem 1: a K_{p+1}-free graph with $\mathrm{e}(T_{n,p}) - t$ edges contains a p-partite subgraph within t edges), applied with $p = 5$ and $n = 26$ where $\mathrm{e}(T_{26,5}) = 270$, the graph H admits a 5-partition whose number of internal (within-part) edges I is at most q. For a partition with part sizes $(n_0, \ldots, n_4)$, set the defect

$$d \;=\; \sum_{i=0}^{4} \binom{n_i}{2} - 55,$$

where $55 = \binom{6}{2} + 4\binom{5}{2}$ is the number of within-part pairs of the balanced Turán partition $(6, 5, 5, 5, 5)$. Counting cross-pairs against $\mathrm{e}(H) = 270 - q$ gives the exact identity

$$\#\{\text{absent cross-pairs}\} \;=\; I + q - d \;=:\; \text{cap}. \tag{1}$$

So each *rung* q is a finite question. Over the partition shapes admissible at that q — those with $\sum_i f(n_i) \leq q$, where $f(n)$ is the least number of internal edges a part of size n must carry so that its own 6-subsets each span at least 4 edges — and over every internal configuration, is there a placement of at most cap absent cross-pairs that keeps H both K_6-free and at least 4 on every 6-set? If the answer is *no* at every admissible (shape, configuration) for a rung, that rung is impossible. Killing the rungs $q = 0, 1, \ldots, 7$ forces $\mathrm{e}(H) \leq 262$, hence $m^* \geq 63$.

- $q \leq 3$: the size-6 part forced by any admissible partition needs $f(6) = 4$ internal edges, but only $I \leq q < 4$ are available. Immediate.
- $q = 4, 5, 6, 7$: each admissible shape and internal configuration is decided by the finite model of Section 3. The shapes are $(6, 5, 5, 5, 5)$ at every rung and, from $q = 6$, also $(7, 5, 5, 5, 4)$; their internal configurations are enumerated up to isomorphism (96 of $(6, 5, 5, 5, 5)$ at $q = 6$; $249 + 52$ across both shapes at $q = 7$). Every one is infeasible.

The lower rungs were also closed by hand: an elementary hole-budget, grid, and pinning argument disposes of $q = 4$ ($m^* \geq 60$) and $q = 5$ ($m^* \geq 61$) without a solver, and the machine reproduces them.

3 The finite decision model and its validation

Each rung-question is a Boolean decision. With the partition and the internal edges fixed, introduce one variable per cross-pair (present or absent) and enforce, for every 6-set, that it spans at least 4 edges and at most 14 (the latter is K_6-freeness), together with the absent-pair budget (1). The encoder is full-specification, with no lemma shortcuts in the trust base.

The model is validated three ways before any infeasibility is trusted.

1. **It reproduces a known value.** The Case-A C_4 configuration of $(6, 5, 5, 5, 5)$ has minimum 34 absent pairs under both an independent checker and the production encoder.
2. **It is not vacuously infeasible.** At a loose budget the model is satisfiable and the returned witness passes an independent verifier.
3. **The isomorphism reduction is exact.** An early version keyed configuration deduplication on a Weisfeiler–Leman hash, which collides on non-isomorphic regular graphs (C_8, $C_4 + C_4$, $C_3 + C_5$); the defect was caught against the validated rung count and corrected to an exact isomorphism-class identity. Without the fix the enumeration silently merges configurations and could report a false closure.

The dense, high-budget configurations at $q = 7$ were confirmed by extended single-instance runs (one required roughly 40 minutes of solver time); all returned infeasible and none feasible.

4 Status, trust base, and reproduction

Proven, unconditional. $m^* \geq 63$ (Theorem 1).

Trust base. (i) Füredi's 2015 stability theorem [2], Theorem 1, a published result whose statement was checked verbatim; and (ii) the full-specification finite model above, validated as in Section 3. No part of the chain depends on any earlier, superseded reduction.

Reproduction. The decision model and the per-rung campaigns are public in the repository linked from the title. Each rung is a self-contained constraint decision; the headline values $a(11)\ldots$ are reproduced by the calibration suite, and the rung enumerations by the multi-shape engine.

5 What $m^* \geq 66$ would still require

The route to the full $r = 5$ case ($m^* \geq 66$) remains open, but it is now sharply localized. Killing rungs $q = 8, 9, 10$ reduces, after the elementary and stability lemmas, to deciding the configurations whose internal graph is *triangle-free and concentrated in one or two parts* ($K_{2,3}$, C_4, C_5, paths). Three independent facts pin down this residue.

- The route is **alive**: the canonical hardest configuration — a star in the 6-part together with a $K_{2,3}$ in one 5-part, at $q = 10$ — has minimum absent-pairs at least 27, exceeding its cap of 20, so it is infeasible, consistent with the conjecture.
- The clique-core hand lemmas (a Case-A argument, an empty-4-part K_6-forcing lemma, and triangle/K_4/K_5 core bounds with machine-exact constants) close the configurations that carry a clique, and an explicit fractional K_6-packing closes the *diffuse* triangle-free configurations.
- However, **fractional and clique methods provably cannot close the concentrated triangle-free residue**: its packing optimum is approximately 13 against a cap of 20, and the gap is inherent. The load-bearing blocking constant $\mathrm{BLOCK}(3, 3, 3, 3) = 18$ is an integer fact whose linear relaxation is only 9.

So $m^* \geq 66$ is reachable as a hybrid — hand lemmas and the packing for the clique-bearing and diffuse families, full-specification constraint search for the concentrated triangle-free core — but that core is the genuine obstruction: each of its configurations needs a heavy individual solver run, and no cheap closed form exists. Proving it is a matter of computation, not of a missing idea.

References

[1] P. Erdős and A. Gyárfás, *A variant of the classical Ramsey problem*, Combinatorica **17** (1997), no. 4, 459–467.

[2] Z. Füredi, *A proof of the stability of extremal graphs, Simonovits' stability from Szemerédi's regularity*, Combinatorica **35** (2015), no. 6, 685–708; arXiv:1501.03129.